

113105011 黃柄耀

DATA SET

DS1530

11. Rating Political Candidates A political scientist is studying the relationship between the voter image of a conservative political candidate and the distance (in kilometers) between the residences of the voter and the candidate. Each of 12 voters rated the candidate on a scale of 1–20.

Voter	Rating	Distance	d_i	d_i^2
1	12	75	3	9
2	7	165	1	1
3	5	300	12	144
4	19	15	1	1
5	17	180	8	64
6	12	240	11	121
7	9	120	4	16
8	18	60	2	4
9	3	230	10	100
10	8	200	9	81
11	15	130	5.5	30.25
12	4	130	5.5	30.25

- Calculate Spearman's rank correlation coefficient r_s .
- Do these data provide sufficient evidence to indicate a negative rank correlation between rating and distance?

$$a. r_s = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)} = 1 - \frac{6 \times 454}{12 \times (12^2 - 1)} \approx -0.581$$

$$b. H_0: \rho_s \geq 0, H_a: \rho_s < 0$$

$$n = 12, \alpha = 0.05, CV = 0.491$$

$$-0.581 \leq -0.491 \Rightarrow \text{Reject } H_0$$

$\Rightarrow$ 選民對候選人的評分與他們之間的距離存在顯著的負相關

12. Competitive Running Is the number of years of competitive running experience related to a runner's distance running performance? The data on nine runners, obtained from a study by Scott Powers and colleagues, are shown in the table:⁸

Runner	Years of Competitive Running		10-Kilometer Finish Time (minutes)		d_i	d_i^2
1	9	1	33.15	1	6	36
2	13	9	33.33	2	7	49
3	5	2.5	33.50	3	0.5	0.25
4	7	6	33.55	4	2	4
5	12	8	33.73	5	3	9
6	6	5	33.86	6	-1	1
7	4	2	33.90	7	-5	25
8	5	3.5	34.15	8	-4.5	20.25
9	3	1	34.90	9	-8	64

- Calculate the rank correlation coefficient between years of competitive running and a runner's finish time in the 10-kilometer race.
- Do the data provide evidence to indicate a significant rank correlation between the two variables? Test using $\alpha = .05$.

$$a, r_s = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)} = 1 - \frac{6 \times 208.5}{9(9^2 - 1)} = -0.1315$$

$$b, H_0: \rho_s = 0, H_a: \rho_s \neq 0$$

$$n = 9, \alpha = 0.05, CV = 0.683$$

$$r_s = -0.1315 \not\leq -0.683 \Rightarrow \text{Reject } H_0$$

$\Rightarrow$ 跑者的競爭性跑步年資與 10 公里完賽時間存在顯著的負相關